Nima Soltani

October 2026

Institute for Research in Fundamental Sciences (IPM)

Shahid Beheshti University

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Other Profiles: ORCID – Google Scholar – ResearchGate – LinkedIn – GitHub – Website

RESEARCH INTERESTS

- Multimodal Reinforcement Learning
- Financial Time Series
- Graph & Hypergraph Networks
- Physics-Informed Neural Networks
- Neural Operators
- Power-System Dynamics
- MEG & EEG Analysis
- Vision–Action
- Deception
- Blockchain / Ethereum

EDUCATION

2023–present
B.Sc. Computer Engineering
Shahid Beheshti University, Tehran. GPA 3.61 / 4.

2020–2023
Diploma in Mathematics & Physics
Sampad (National Organization for Development of Exceptional Talents), Zanjan. GPA 3.88 / 4. Ranked 254th nationwide (top 0.2%), Iranian National University Entrance Exam, 2023.

2017–2020
Middle school
Sampad, Zanjan. GPA 20 / 20.

RESEARCH APPOINTMENTS

2023–present
Research Assistant
Institute for Research in Fundamental Sciences (IPM), with Dr. Dara Rahmati. Deep reinforcement learning and multimodal models for market modelling and automated trading (transformer and diffusion-policy agents; fused xLSTM–PPO; graph and hypergraph models of market structure); physics-informed Kolmogorov–Arnold networks for PDEs; physics-informed operator networks for power-system swing dynamics.

2024–present
Research Assistant
Shahid Beheshti University · Blockchain Systems, with Dr. Maedeh Mosharraf. Blockchain architecture and Ethereum protocol design, Solidity smart contracts, consensus, on-chain transaction modelling, and security analysis of cryptocurrency and DeFi infrastructure.

2026–present
Research Assistant
Shahid Beheshti University · Computational Neuroscience, with Dr. Narjes Soltani. MEG preprocessing, source localization, and neural decoding, with spectral and time–frequency analysis related to visual perception and cognition.

PUBLICATIONS

- *[Under review]* **N. Soltani**, A. Sadr, A. Pakniyat, K. Pakzad, and D. Rahmati. **Automated Stock Trading Using Deep Reinforcement Learning and Fused xLSTM Networks**. *Applied Soft Computing*, Elsevier. Preprint on SSRN.
- *[Under review]* **N. Soltani**, A. Sadr, and D. Rahmati. **Robust Multimodal Portfolio Trading Using Transformer-Based PPO: Technical–Fundamental Fusion with Conservative On-Policy Diffusion**. *Knowledge-Based Systems*, Elsevier.
- *[Under review]* **N. Soltani**, A. Sadr, and D. Rahmati. **Physics-Informed Deep Operator Networks for Classical Swing Dynamics in Single-Machine Infinite-Bus Systems**. *IEEE Transactions on Power Systems*, IEEE.

- *[Under review]* **N. Soltani**, M. Eidi, A. Sadr, and D. Rahmati. **A Lightweight xLSTM-BLS Method with Non-linear Feature Fusion for Accurate Epileptic Seizure Detection Using EEG Signals**. *Biomedical Signal Processing and Control*, Elsevier.
- *[Preprint]* A. Sadr, **N. Soltani**, V. Moghtadaiee, A. Pakniyat, D. Rahmati, and S. Gorgin. **PG-KINN: A Physics-Informed Petrov–Galerkin Kolmogorov–Arnold Network for Solving Forward and Inverse PDEs**. arXiv preprint.

TEACHING ASSISTANTSHIPS

Since 2024 at Sharif University of Technology, Amirkabir University of Technology, and Shahid Beheshti University.

Sharif University of Technology

Data Mining — Dr. Mohammad Khalooei — Sep 2026–Feb 2027
Compiler — Ms. Samane Hosseinmardi — Sep 2026–Feb 2027
Machine Learning Systems Engineering — Dr. Maryam Ramezani — Sep 2026–Feb 2027
Logic Circuits — Dr. Laleh Arshadi — Sep 2026–Feb 2027
Logic Circuits — Dr. Shahin Hessabi — Feb–Jul 2026
Logic Circuits — Dr. Laleh Arshadi — Feb–Jul 2026
Scientific and Technical Presentation — Dr. Laleh Arshadi — Feb–Jul 2026
Computer Architecture — Dr. Laleh Arshadi — Sep 2025–Feb 2026
Logic Circuits — Dr. Laleh Arshadi — Sep 2025–Feb 2026
Fundamentals of Programming — Mr. Mohammad Nazari — Sep 2025–Feb 2026
Computer Architecture — Dr. Laleh Arshadi — Jan–Jun 2025
Fundamentals of Programming — Ms. Marziyeh Sadri — Jan–Jun 2025
Fundamentals of Programming — Mr. Mohammad Zarei — Jan–Jun 2025
Fundamentals of Programming — Mr. Mohammad Nazari — Sep 2024–Jan 2025

Amirkabir University of Technology

Data Mining — Dr. Behnam Yousefimehr — Sep 2026–Feb 2027

Shahid Beheshti University

Principles of Design and Analysis of Experiments (Graduate; Head TA) — Dr. Narjes Soltani Dehaghani Sep 2026–Feb 2027
Signals and Systems and Digital Control (Graduate; Head TA) — Dr. Zeinab PourbafaraniSep 2026–Feb 2027
Database — Dr. Hassan Haghighi — Sep 2026–Feb 2027
Engineering Probability and Statistics (Head TA) — Dr. Narjes Soltani Dehaghani — Sep 2026–Feb 2027
Blockchain Technology (Graduate) — Dr. Maedeh Mosharraf Dehkordi — Sep 2026–Feb 2027
Blockchain Technology (Graduate) — Dr. Maedeh Mosharraf Dehkordi — Feb–Jul 2026
Linear Algebra (Head TA) — Dr. Farokhlagha Moazami Goodarzi — Feb–Jul 2026
Fundamentals of Data Mining (Head TA) — Dr. Ahmad-Ali Abin — Feb–Jul 2026
Advanced Programming — Dr. Mojtaba Vahidi Asl — Feb–Jul 2026
Data Structures and Algorithms — Dr. Ahmad-Ali Abin — Sep 2025–Feb 2026
Fundamentals of Programming — Dr. Zeinab Pourbafarani — Sep 2025–Feb 2026
Fundamentals of Programming (Head TA) — Dr. Mahdi Mojahedi — Sep 2025–Feb 2026
Computer Architecture — Dr. Dara Rahmati — Sep 2025–Feb 2026
Formal Languages and Automata Theory — Dr. Ramak Ghavami Zadeh — Jan–Jun 2025
Computer Architecture — Dr. Razieh Salarifard — Jan–Jun 2025
Advanced Programming — Dr. Sadegh Aliakbari — Jan–Jun 2025
Advanced Programming — Dr. Mojtaba Vahidi Asl — Jan–Jun 2025
Fundamentals of Programming — Dr. Monireh Abdoos — Jan–Jun 2025
Applied Computer Skills — Dr. Hamed Malek — Sep 2024–Jan 2025

VOLUNTARILY ACTIVITIES

Workshops

- **Fundamentals of Programming.** Pointers, dynamic memory, and recursion in C: memory layout, `malloc/free`, dangling pointers and leaks; recursion from base cases through the call stack, with backtracking and divide-and-conquer exercises.
- **Logic Circuits.** Combinational and sequential design: Boolean minimisation, building blocks, flip-flops, and finite-state machines, with designs checked in a simulation tool.
- **Linear Algebra.** SVD, PCA, and numerical methods in Python: decompositions in NumPy, PCA on real data, and why numerical stability matters in practice.
- **Advanced Programming.** Java OOP (classes, interfaces, polymorphism) and a small 2D Unity game in C# organised with object-oriented patterns.
- **Data Structures and Algorithms.** Graph traversal and dynamic programming: BFS, DFS, and shortest paths; turning problems into recurrences solved with memoisation or tabulation.
- **Data Mining.** Practical sessions with scikit-learn on preprocessing, feature engineering, clustering, and classification, with attention to evaluation and data leakage.
- **Machine Learning Systems Engineering.** Reproducible ML pipelines: experiment tracking, configuration management, versioned data and models, and basics of packaging and deploying a trained model.
- **Principles of Design and Analysis of Experiments.** Experiment design and simulation in Python and R: randomisation, blocking, and factorial designs; simulation for effect sizes, power, and ANOVA.
- **Blockchain Technology.** Ethereum smart contracts and security: writing, testing, and deploying Solidity contracts; common vulnerabilities (reentrancy, access control) and how audits find them.
- **Computer Architecture.** Instruction set architecture and MIPS assembly: instruction formats, addressing modes, register conventions, and hand-written assembly for loops, procedures, and the stack.

Service

- **Volunteer mentor** for first-year computer engineering students at Shahid Beheshti University, helping with coursework, study habits, and settling into university life.
- **Secretariat member**, 1st International Conference on Artificial Intelligence, Tehran, February 2025.
- **Problem-solving and review sessions** before midterms and finals, working through exam-style problems with students.
- **Assignment and project design**: programming assignments, projects, test cases, and grading rubrics, plus course materials (worked solutions, slides, short guides) and project mentoring with detailed code review.

SELECTED COURSE PROJECTS

- **Adversarial Game Agent** — Artificial Intelligence
  Minimax with alpha–beta pruning and heuristic evaluation.
- **SVD Recommender** — Linear Algebra
  SVD/PCA recommender evaluated by RMSE and MAE.
- **Virtual Memory Simulator** — Operating Systems
  LRU and Second Chance page-replacement comparison.
- **SBUMIPS Processor** — Computer Architecture
  Pipelined MIPS-like processor in Verilog (ModelSim).
- **Survival Game in Unity** — Advanced Programming
  2D game with an object-oriented architecture.
- **File Compression Tool** — Fundamentals of Programming

Custom encoding with attention to time and memory.

SKILLS

Programming	Python, C, C++, C#, Java, MATLAB, R, SQL, Verilog
Deep learning	PyTorch, PyTorch Lightning, JAX, Hugging Face, scikit-learn, ONNX, OpenCV
RL & sequences	PPO, xLSTM, diffusion policies, Stable-Baselines3, Gymnasium, FinRL, transformers, Mamba/SSM
Graphs & physics	PINNs, KANs, PyTorch Geometric, DGL, hypergraph networks, DeepXDE, DeepONet, FNO
Signals	MNE-Python, FieldTrip, Brainstorm, EEGLAB, source localization, time–frequency analysis, neural decoding
Data & finance	SQL, NumPy, pandas, SciPy, Matplotlib, Seaborn, statsmodels, XGBoost, LightGBM, CatBoost, Dask, Microsoft Excel, Power BI, yfinance, pandas-ta, FinBERT
Blockchain	Solidity, Ethereum, Hardhat, Foundry, web3.py, ethers.js, OpenZeppelin, Slither
Infrastructure	Git, Linux, Docker, CUDA, SLURM/HPC, Weights & Biases, MLflow, TensorBoard, Hydra, Optuna, LaTeX/Overleaf, Zotero, ModelSim, Unity

HONORS, TRAINING, AND MEETINGS

Distinctions & training

- Ranked 254th nationwide (top 0.2%), Iranian National University Entrance Exam, 2023.
- NeuroCONNECT Summer School in Computational Neuroscience (CONNECT), Aug 2026.
- Advanced Algorithmic Thinking and Data Structures Certificate (Quera).
- Academic Article Writing Workshop, Institute for Advanced Studies in Basic Sciences.
- Serious Game Design with a Cognitive Approach, workshop, Shahid Beheshti University.
- ICDL Certification, Technical and Vocational Training Organization.

Conferences & memberships

- Participant, International Interdisciplinary Conference on Artificial Intelligence (IICAI), Shahid Beheshti University / IEEE, May 2026.
- Participant, 9th Conference on Advances in Enterprise Architecture of Iran, Shahid Beheshti University, Nov 2025.
- ISCISC 2025 specialized workshops (BIOS security, fault injection, code readout, smart-card authentication, AI for cybersecurity), Oct 2025.
- Prompt Engineering for Large Language Models, workshop, 1st ICAI, Feb 2025.
- Member, Iranian Bioinformatics Society (IBIS).

LANGUAGES

Persian	Native
Turkish	Native
English	Professional working proficiency